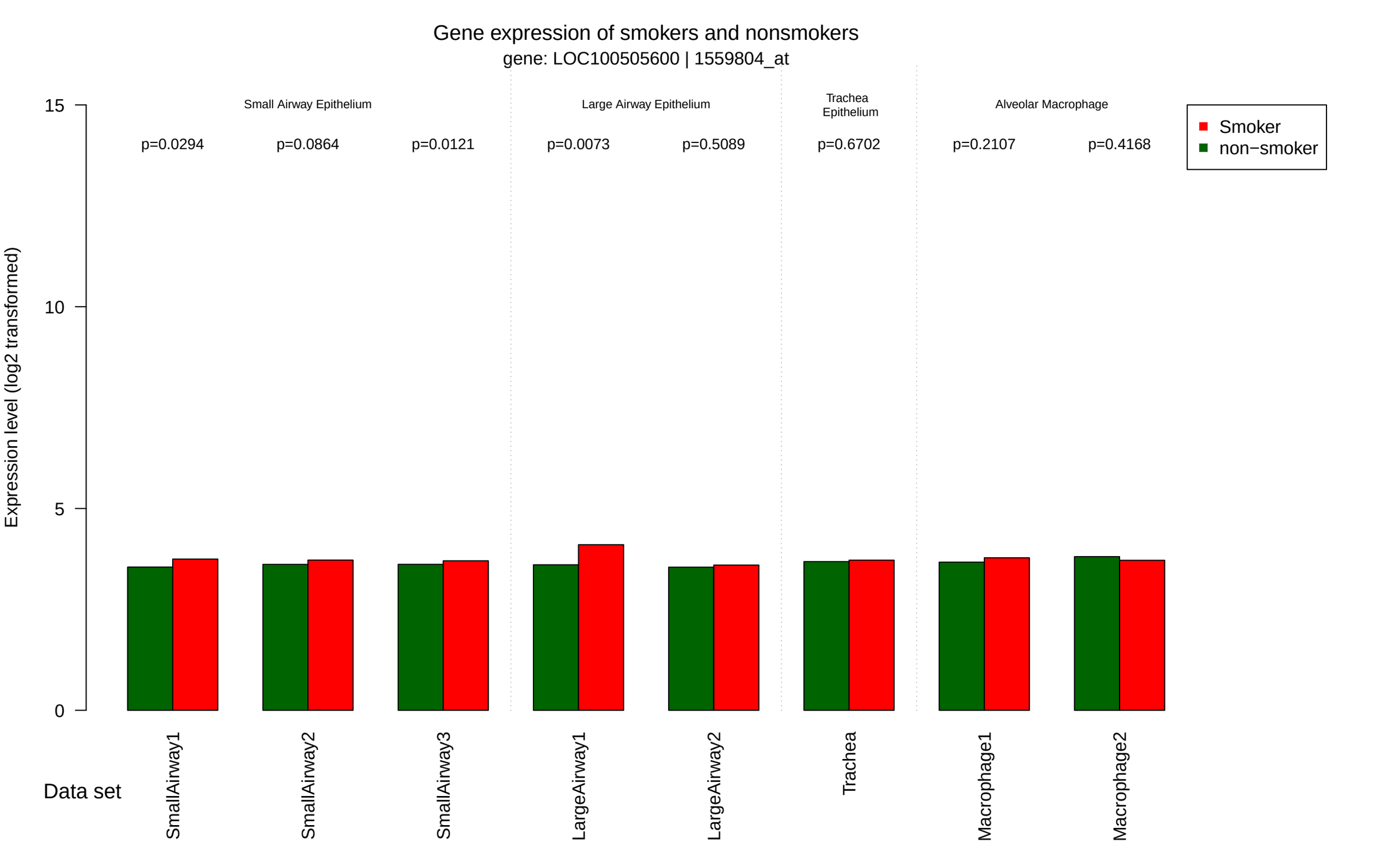
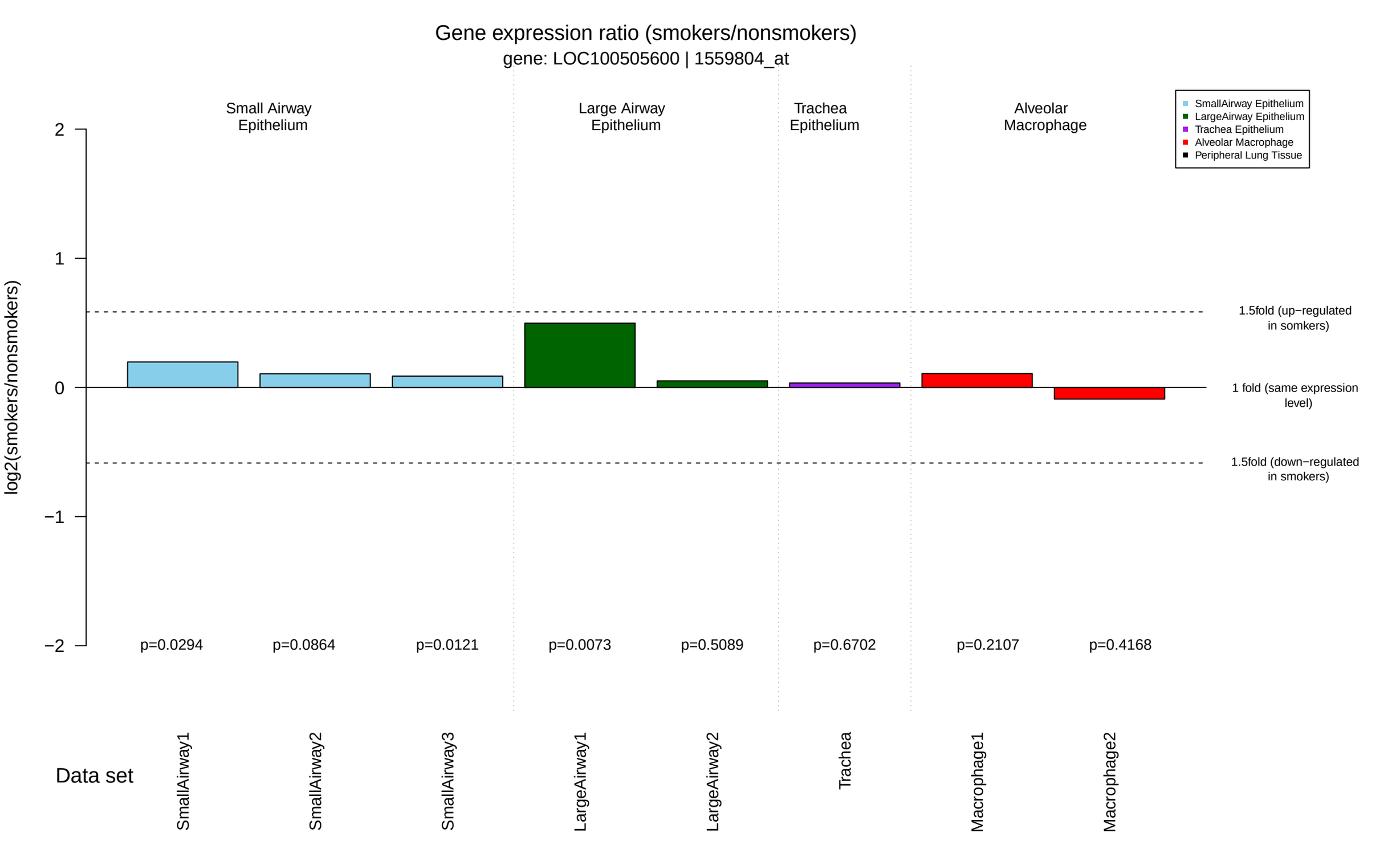
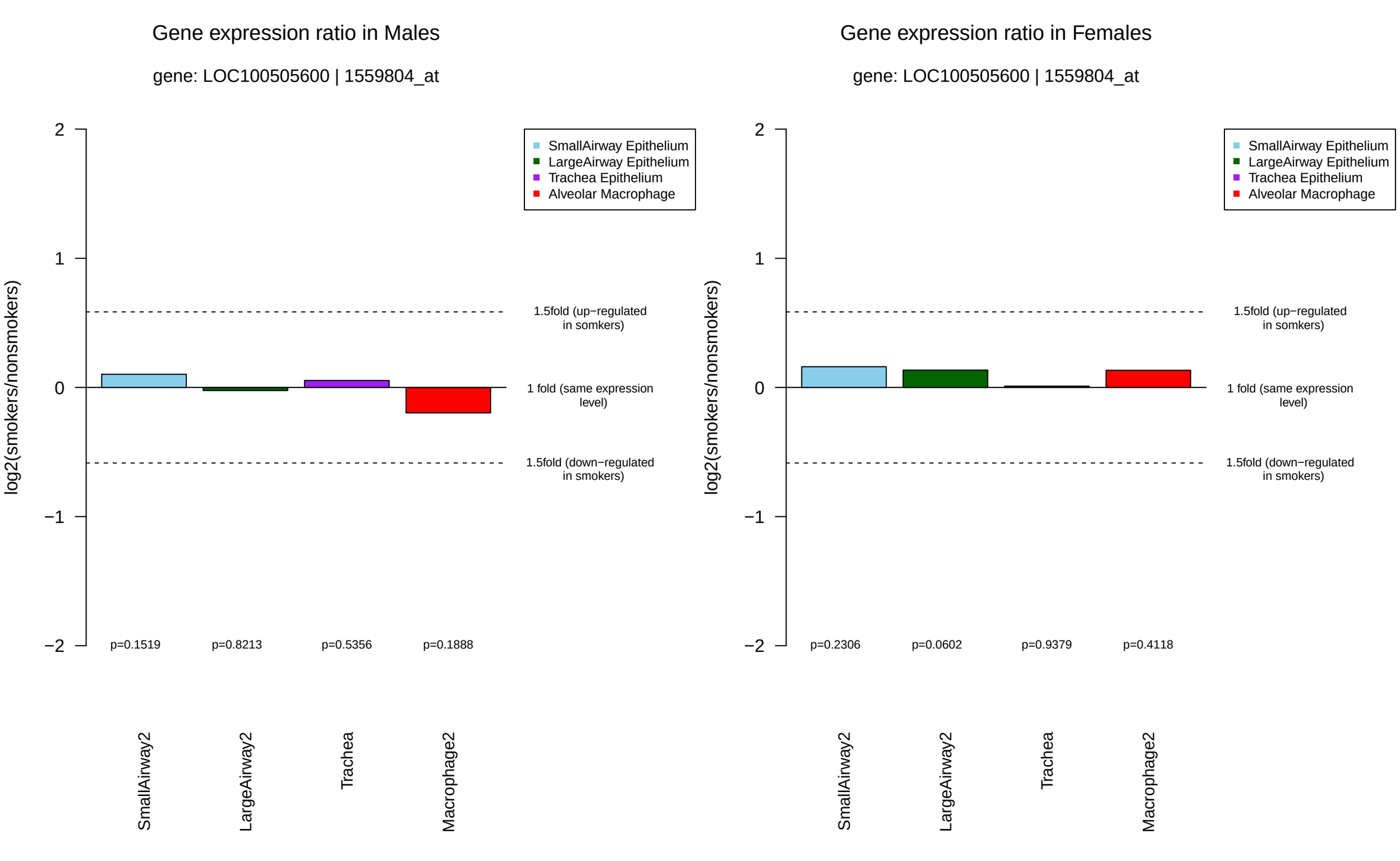
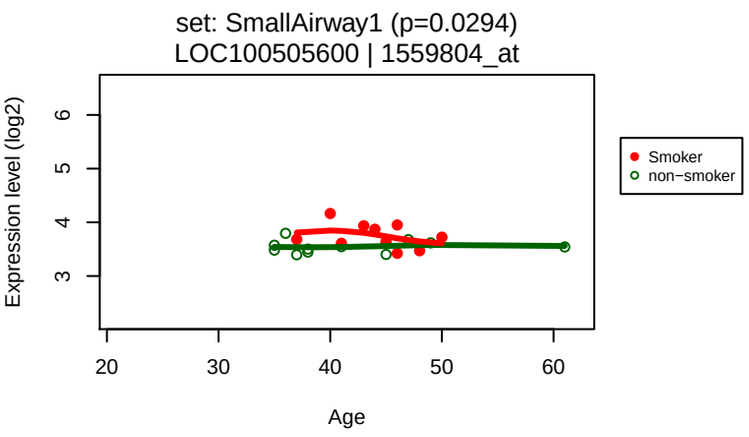
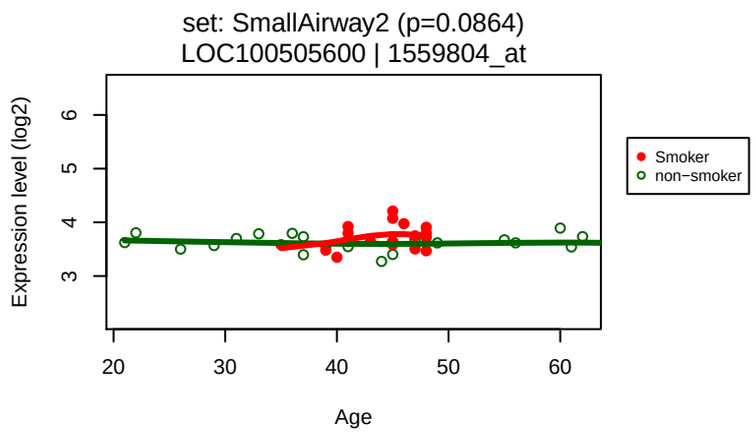
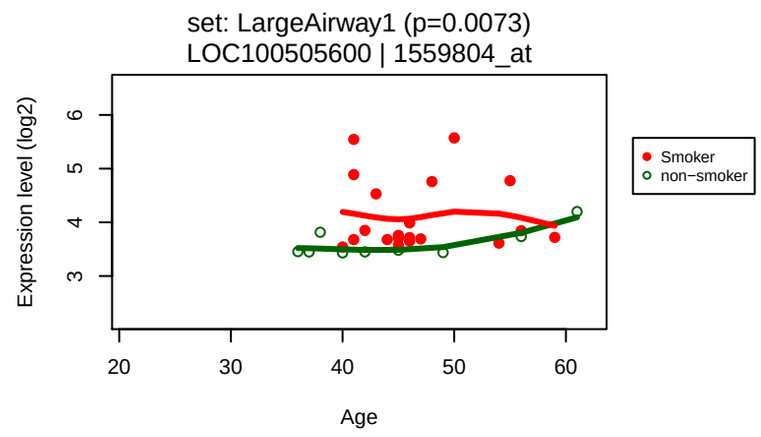

Gene expression of smokers and nonsmokers
gene: LOC100505600 | 1559804_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0294
p=0.0864
p=0.0121
p=0.0073
p=0.5089
p=0.6702
p=0.2107
p=0.4168
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

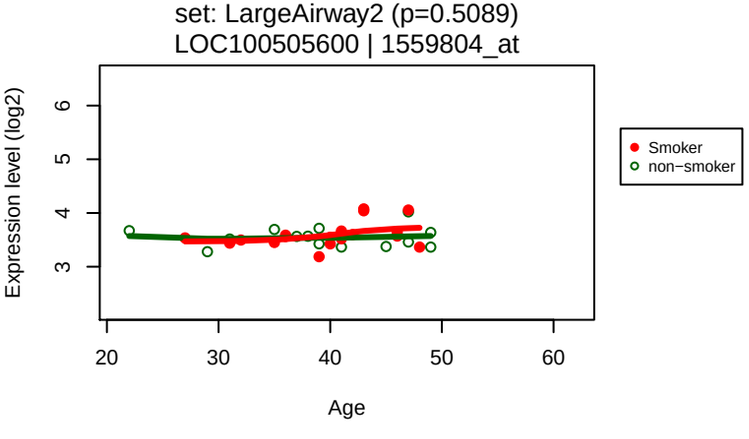

Gene expression ratio (smokers/nonsmokers)
gene: LOC100505600 | 1559804_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0294
p=0.0864
p=0.0121
p=0.0073
p=0.5089
p=0.6702
p=0.2107
p=0.4168
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

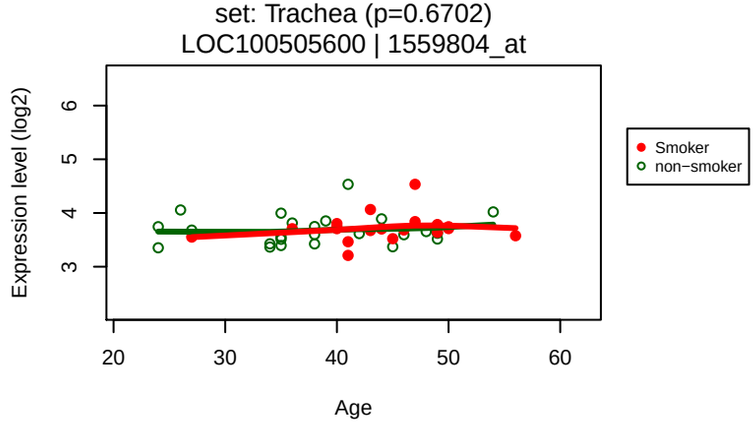

Gene expression ratio in Males
gene: LOC100505600 | 1559804_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1519
p=0.8213
p=0.5356
p=0.1888
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC100505600 | 1559804_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2306
p=0.0602
p=0.9379
p=0.4118
SmallAirway2
LargeAirway2
Trachea
Macrophage2

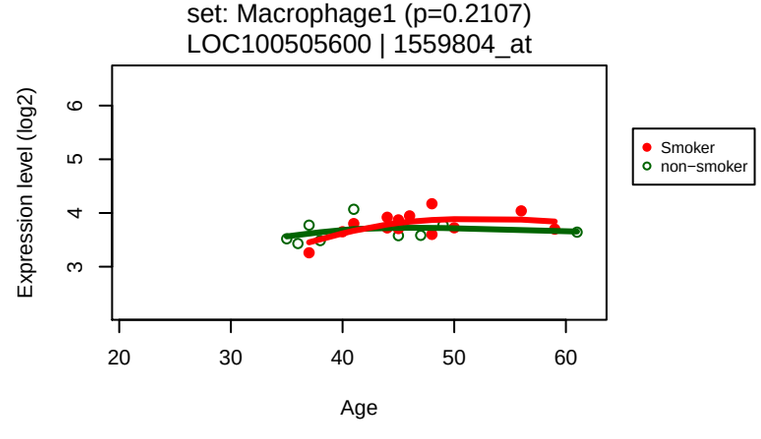

set: SmallAirway1 (p=0.0294)
LOC100505600 | 1559804_at
Expression level (log2)
Age
Smoker
non−smoker

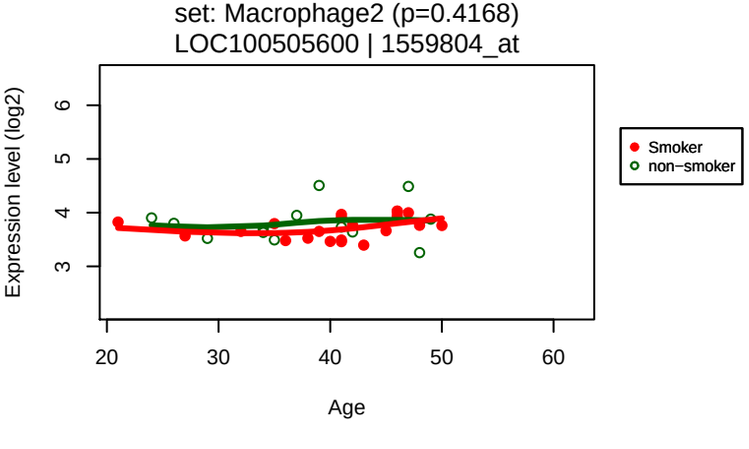

set: SmallAirway2 (p=0.0864)
LOC100505600 | 1559804_at
Expression level (log2)
Age
Smoker
non−smoker

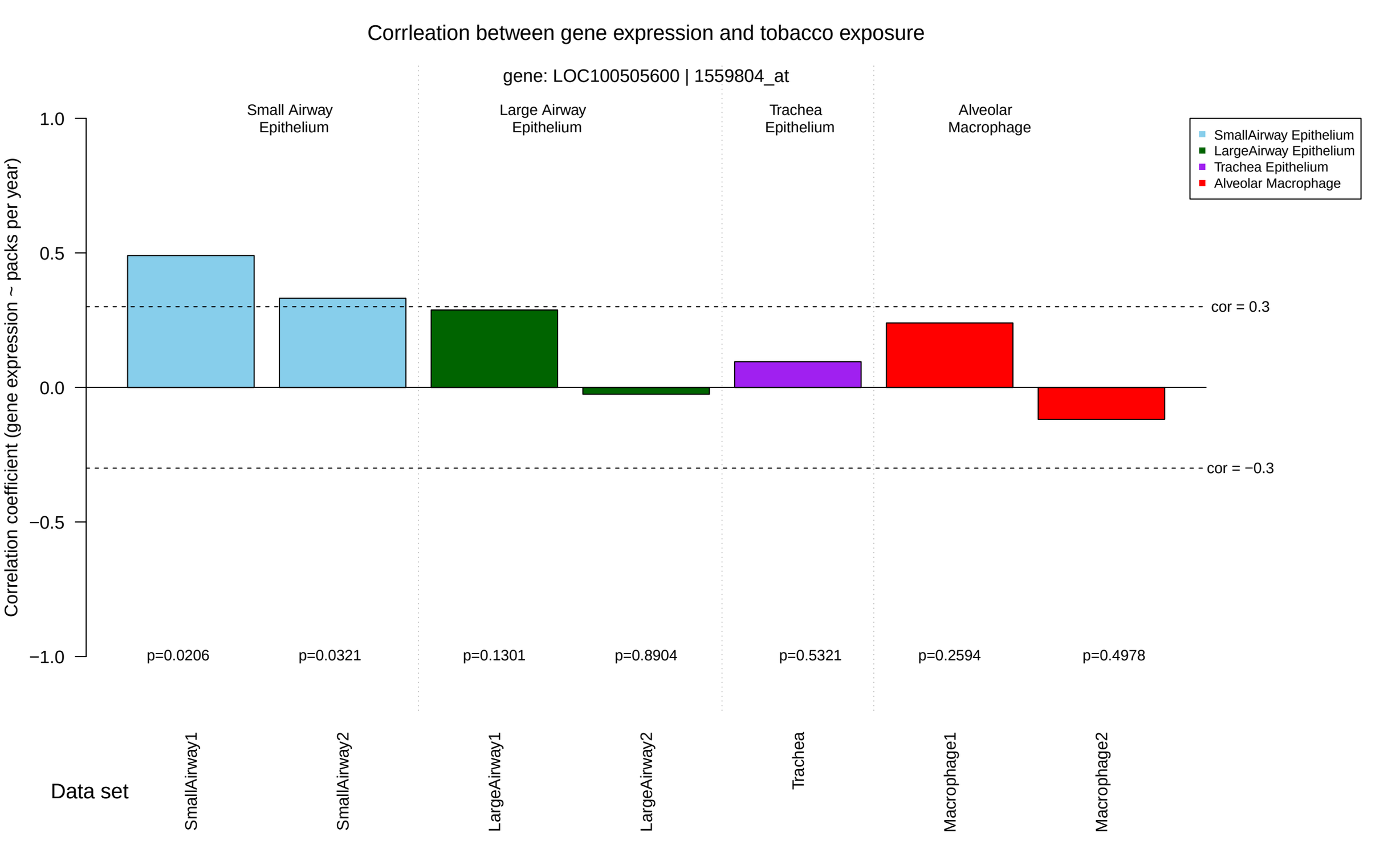

set: LargeAirway1 (p=0.0073)
LOC100505600 | 1559804_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5089)
LOC100505600 | 1559804_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6702)
LOC100505600 | 1559804_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2107)
LOC100505600 | 1559804_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4168)
LOC100505600 | 1559804_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC100505600 | 1559804_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0206
p=0.0321
p=0.1301
p=0.8904
p=0.5321
p=0.2594
p=0.4978
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2